

PHASE 4

PROJECT DEVELOPMENT

4.1 DATA MIGRATION, TESTING & SECURITY

This phase covers the essential activities of migrating lease-related data into Salesforce, thorough testing of the developed features, and ensuring security measures are properly implemented. The objective is to guarantee data accuracy, system reliability, and protection of sensitive information before the system goes live.

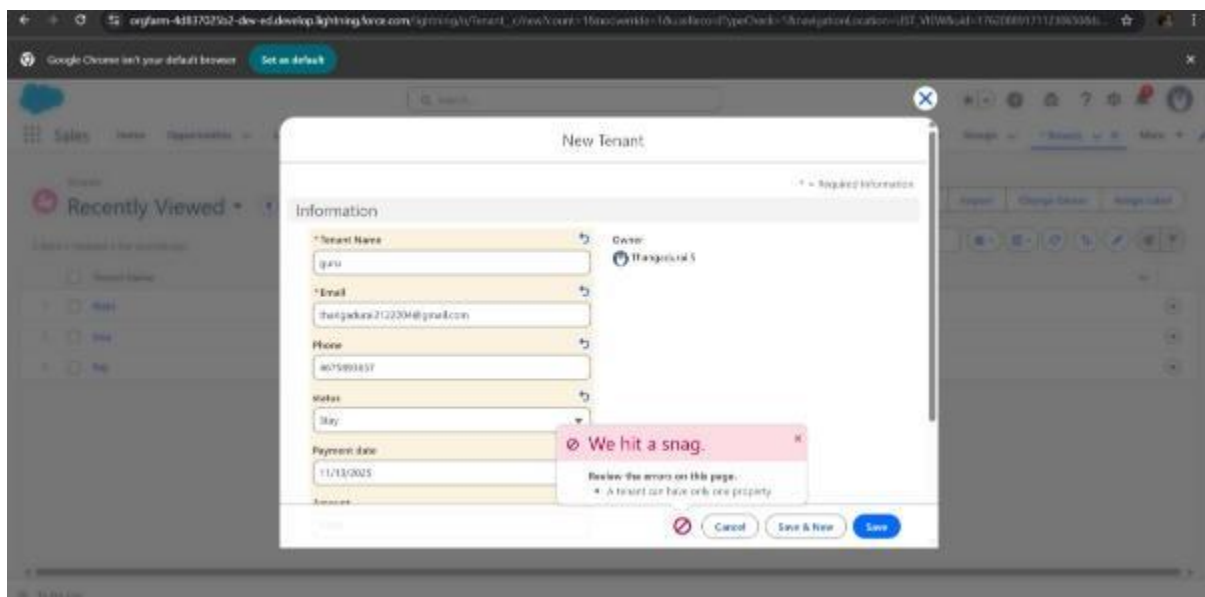
4.2 APEX TRIGGER TESTING

A custom Apex trigger was developed to enforce the rule that each property can be assigned to only one tenant at a time.

To test this:

- A test class was created simulating insertion of tenant records.
- It confirmed successful creation when a property was free.
- It verified the trigger correctly throws an error if a property is already assigned, preventing data inconsistencies.

Test logs and results demonstrated the trigger's effectiveness in maintaining data integrity.

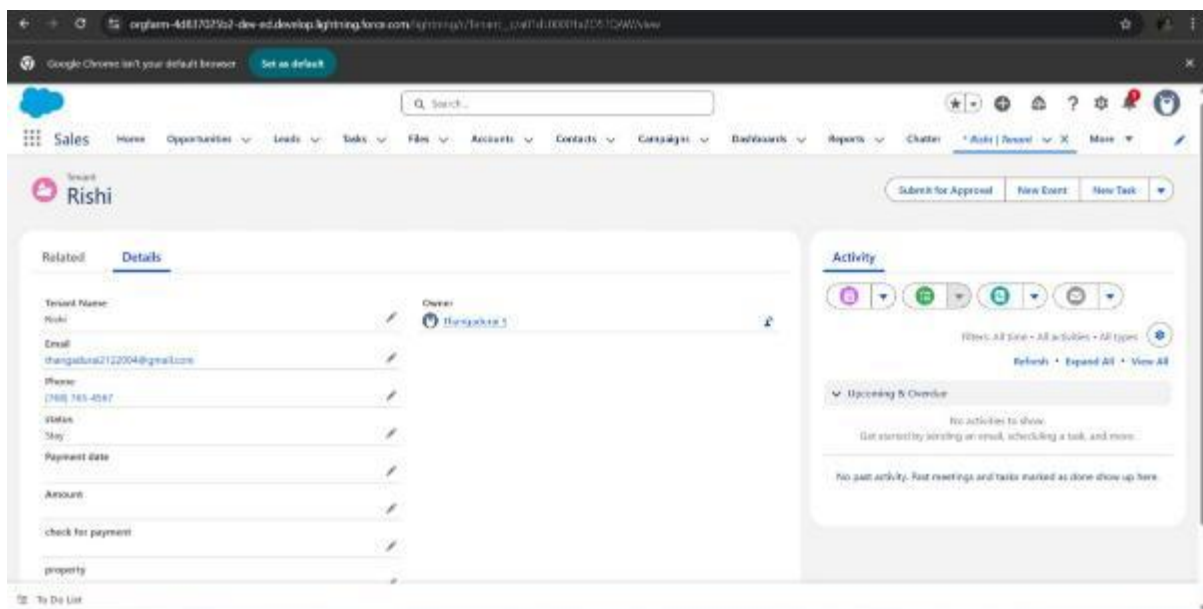


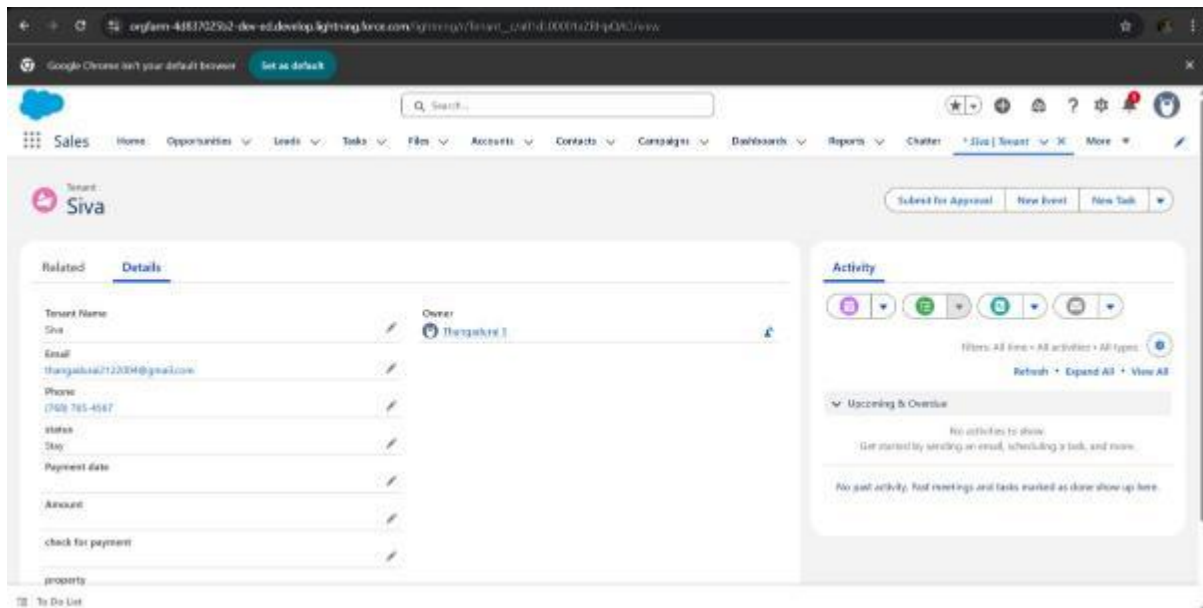
4.3 APPROVAL PROCESS TESTING

The tenant leave request approval workflow was tested through multiple scenarios:

- Submission of leave requests by tenants triggered the approval process.
- Approvers received real-time notifications and emails.
- Both approval and rejection paths were tested to ensure that tenant status updated accordingly, and the proper email templates were sent to notify tenants.

Screenshots of submission forms, approval screens, and notification emails are documented as evidence of successful workflow execution.





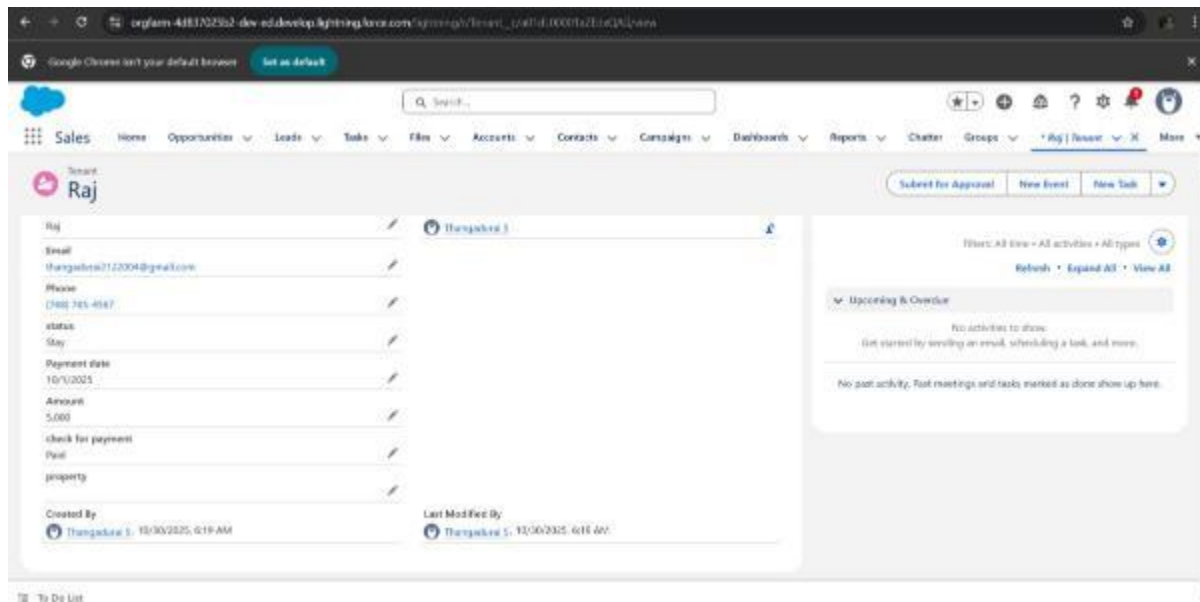
4.4 FLOW TESTING

The system includes a record-triggered flow that sends payment confirmation emails automatically when a tenant's payment status changes to "paid."

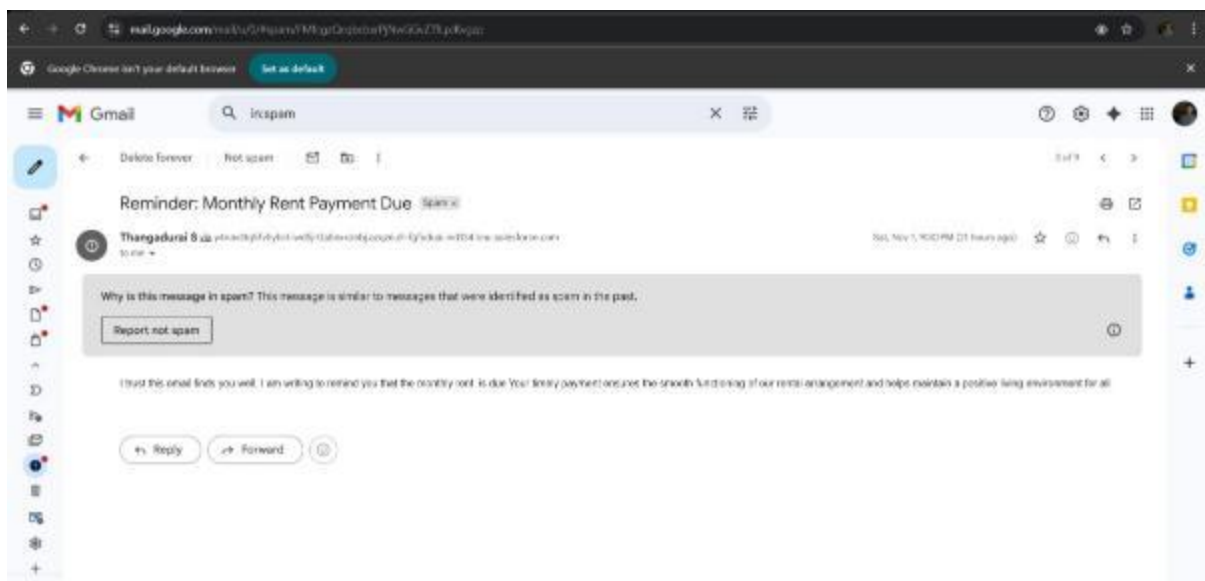
Testing involved,

- Updating payment records to "paid" status.
- Monitoring flow execution in Salesforce.
- Confirming that tenants received correctly formatted confirmation emails.

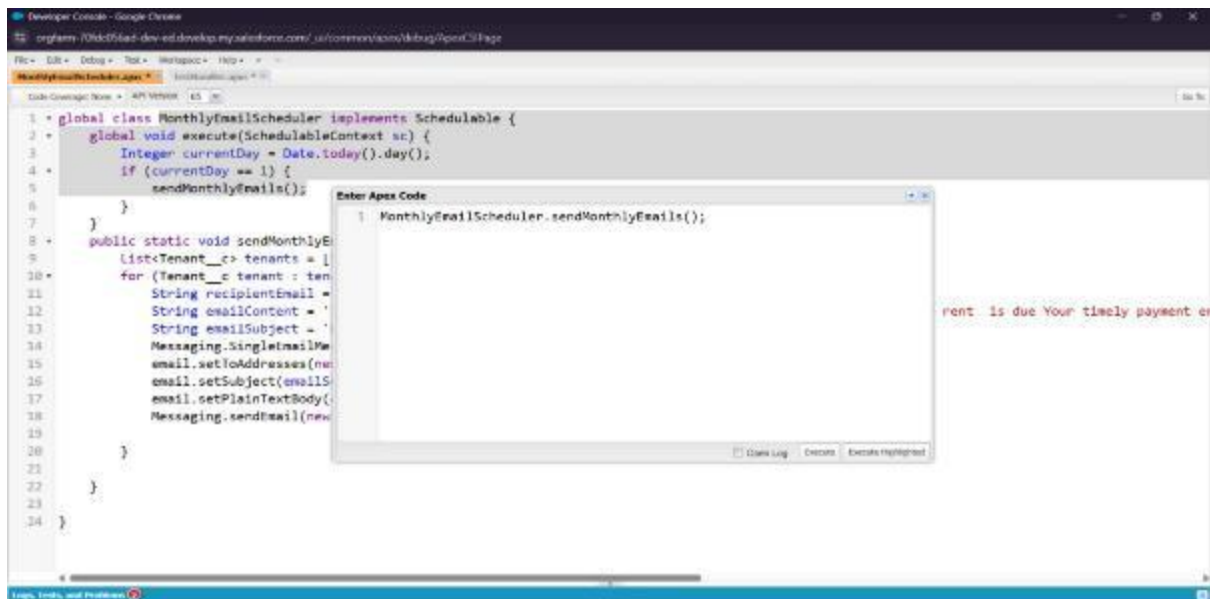
This automated communication improves tenant engagement and reduces manual workload.



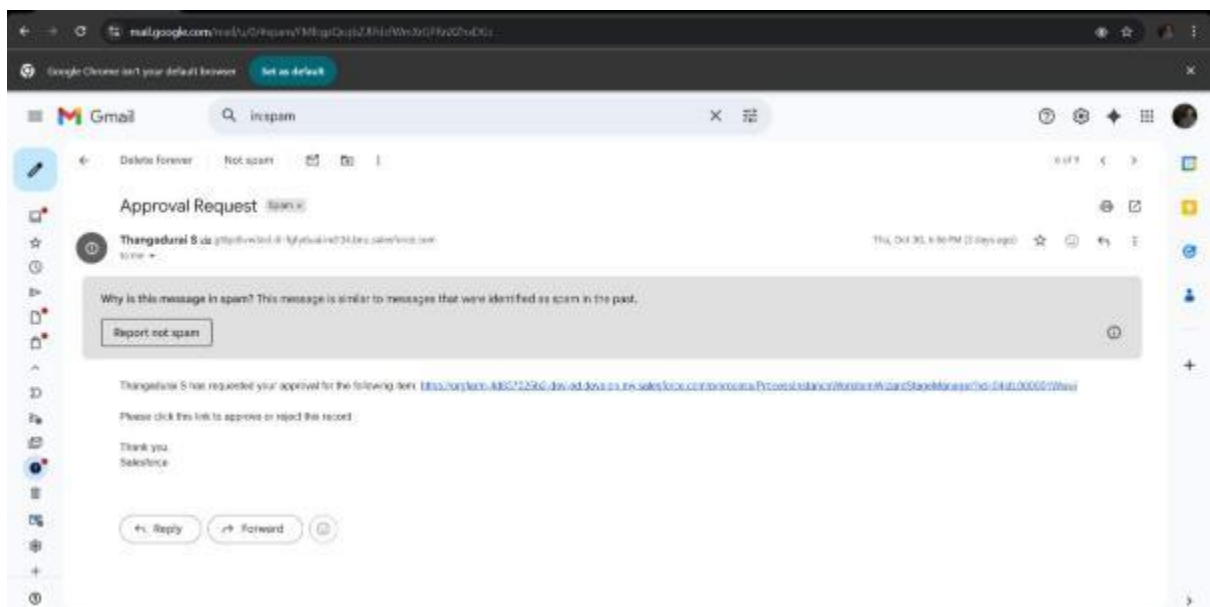
A confirmation mail received to the tenant about the payment of the monthly rent.

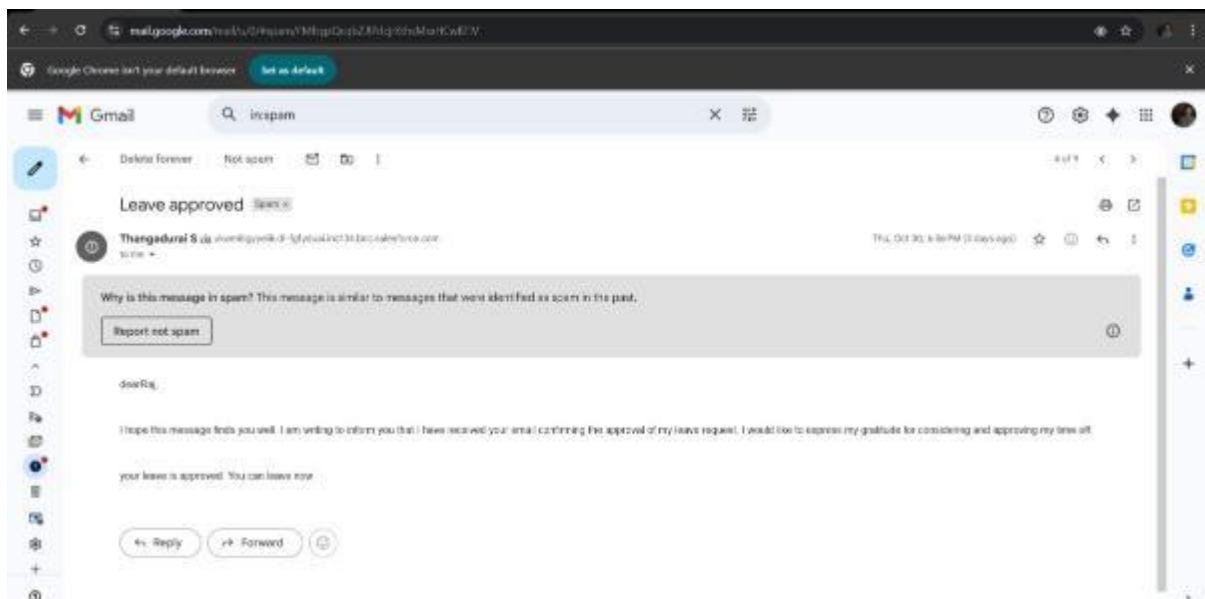
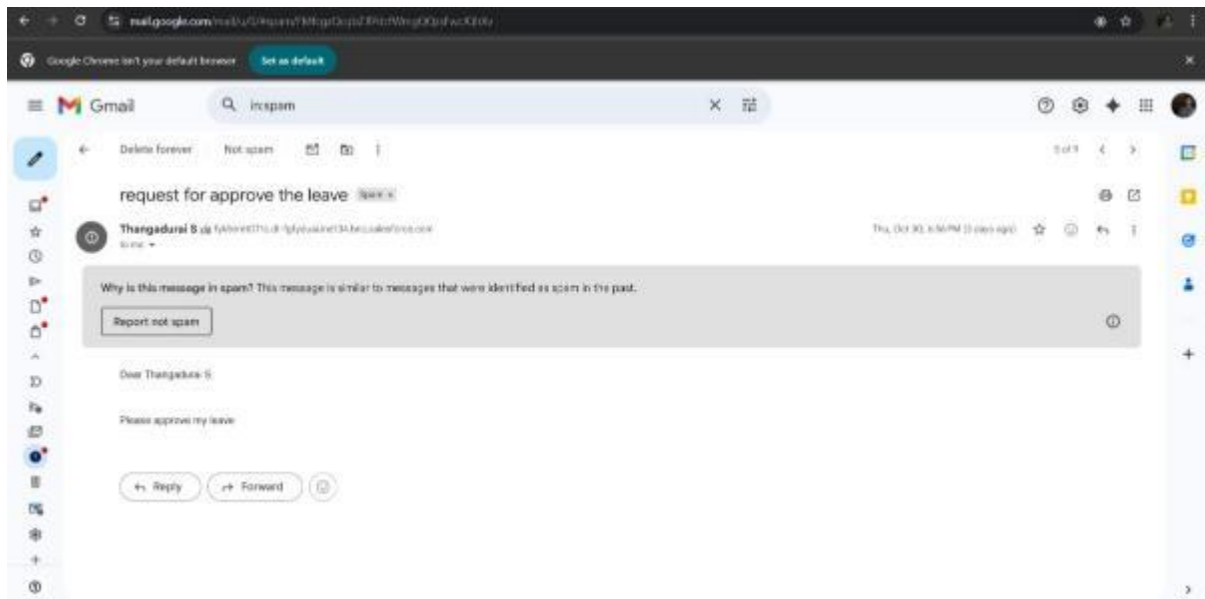


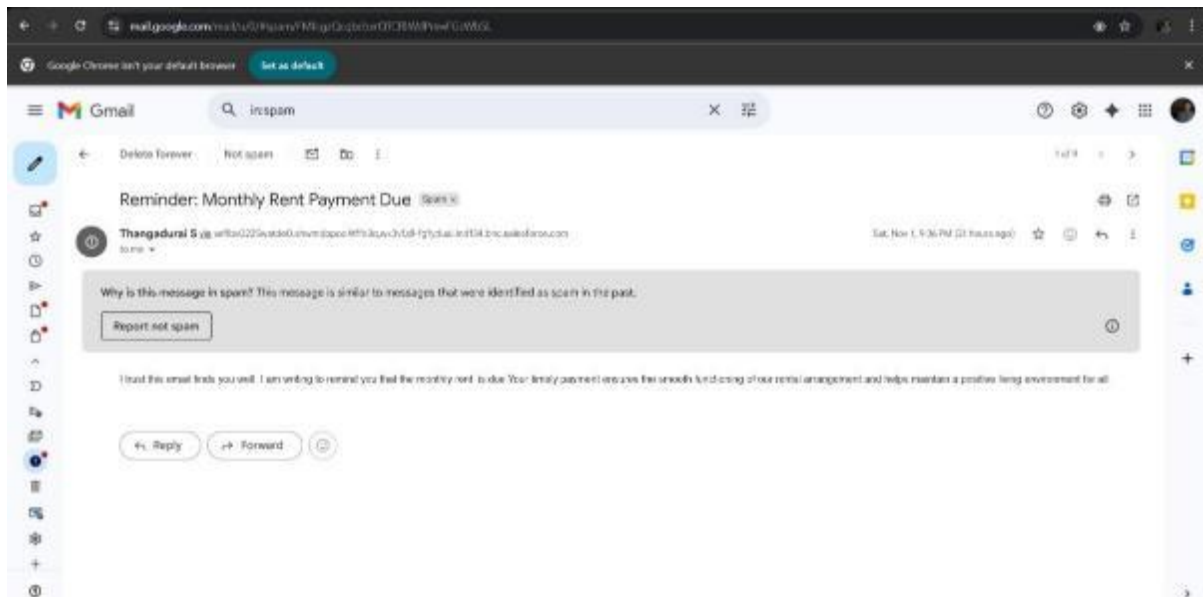
4.5 SCHEDULED APEX TESTING



The execution of the `MonthlyEmailScheduler` class.





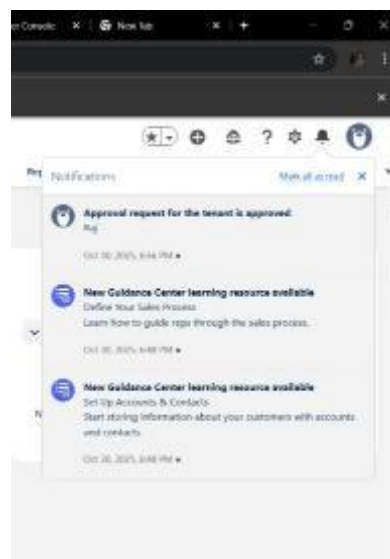


4.6 DOCUMENTATION AND EVIDENCE

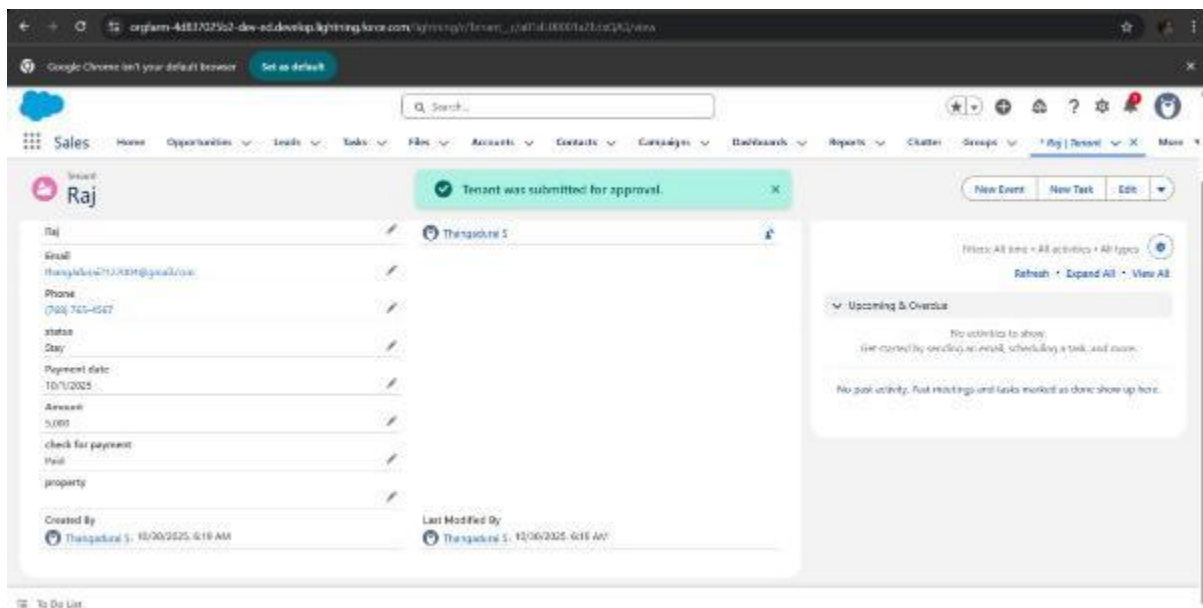
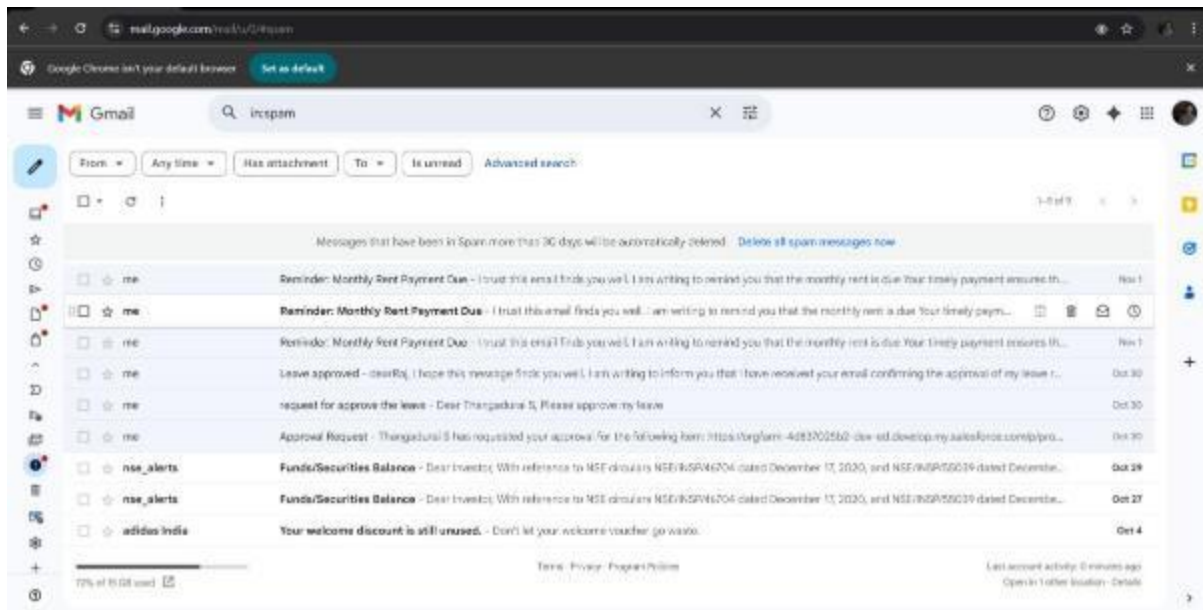
- Detailed test cases were created covering all implemented features.
- Input data, execution steps, and expected outcomes were recorded for each test.
- Screenshots captured actual results to verify correct functionality.
- Documentation ensures transparency and aids future maintenance.
- Provides confidence in system readiness for deployment.

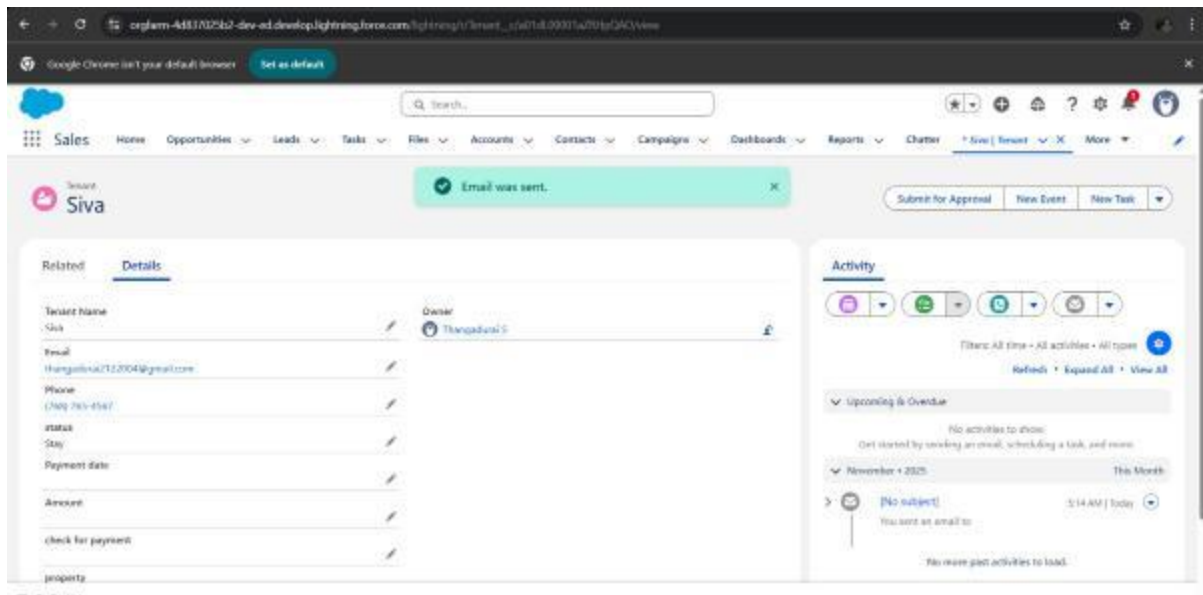
The execution of the Apex class MonthlyEmailScheduler. The following figure shows the execution logs of the Apex class.

The notifications are received to the user.



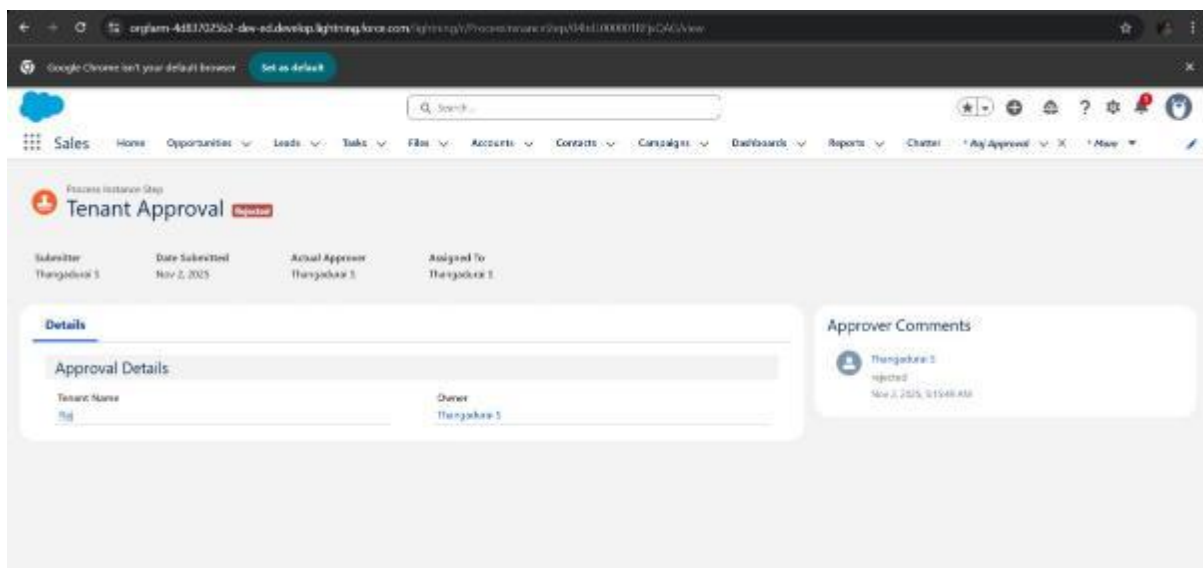
Emails received by the tenant sent by the user or owner through automated process.



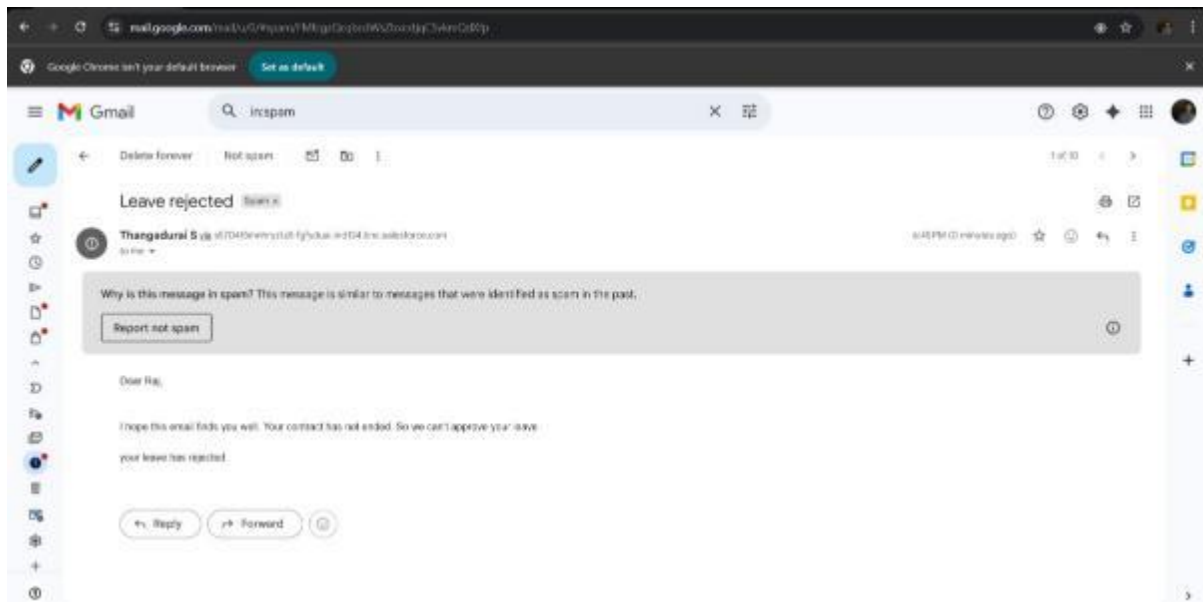


The above figure determines the approval page for the owner. The owner can approve, reject or reassign the tenant request.

The approver can accept or reject the user request.



The approval or rejection email is received to the tenant.



4.7 DEPLOYMENT STRATEGY

To ensure a smooth transition from development to production, a well-defined deployment strategy has been implemented. The primary method used for deployment is Salesforce Change Sets, allowing for the structured movement of metadata between environments. This approach reduces risks, ensures stability, and maintains system integrity.

4.7.1 KEY COMPONENTS OF DEPLOYMENT

- Developer Sandbox Usage:

All development and testing activities are conducted in Salesforce Developer Edition or Sandbox environments. This isolates testing from the live system and allows teams to experiment and validate without affecting end users.

OUTBOUND CHANGE SETS

These are created in the Sandbox environment and include all the necessary components such as:

- Custom Objects and Fields
- Validation Rules o Classes and Triggers
- Process Builders and Flows
- Lightning Pages and Components Apex

INBOUND CHANGE SETS

The packaged Change Sets are received in the Production Org and reviewed. Once verified, they are deployed after proper validation and testing, ensuring that only stable and approved features go live.

- Pre-deployment Testing:

Before any deployment, a complete testing cycle is conducted in a **Full Sandbox** environment.

This includes:

- Functional Testing
- Integration Testing
- Regression Testing

This helps identify potential bugs, configuration issues, or deployment errors early in the process.

4.8 ADVANCED DEPLOYMENT TOOLS

For larger projects or distributed teams, additional tools are considered for improved control and automation:

- SFDX CLI (Salesforce DX) for script-based deployment and continuous integration
- Git for version control and team collaboration

This deployment approach ensures consistency, minimizes risks, and supports scalable growth in the future.

4.9 SYSTEM MAINTENANCE & MONITORING

Post-deployment, the system needs to be actively monitored and maintained to ensure continued performance and data reliability. A structured maintenance plan has been put in place:

KEY ACTIVITIES:

- **Regular Data Backups:** Scheduled backups are configured to safeguard critical information such as lease contracts, tenant records, payment data, and related documents. Backups ensure business continuity in case of data loss or system failure.

- **Performance Monitoring:** Tools provided by Salesforce are used to continuously monitor system performance:

- Debug Logs to track system executions and identify performance bottlenecks
- Setup Audit Trail to monitor configuration changes and user actions
- Salesforce Health Check to assess security settings and overall system actions.

- **User Feedback Collection:** Regular feedback is gathered from end-users (property managers, admins, tenants) to identify UX/UI improvements and minor bugs. Feedback is logged and analyzed to drive iterative enhancements.

- **Scheduled Enhancements & Upgrades:** Based on usage patterns and evolving business needs, new features or process optimizations are scheduled periodically.

These may include:

- New report types
- Enhanced dashboards
- Additional automation (e.g., lease expiry followups)
- Workflow optimizations

- **Security & Permission Reviews:** As teams grow or roles change, it's crucial to ensure access levels are aligned.

Periodic reviews are conducted for,

- Role hierarchy
- Profile settings
- Permission sets and sharing rules

These reviews ensure that users only access the data they're authorized to, maintaining trust and data protection compliance.

4.10 DOCUMENTATION & TROUBLESHOOTING

Proper documentation is essential for smooth adoption, ongoing support, and future scalability. Two levels of documentation have been created:

4.10.1 USER DOCUMENTATION

Designed for end-users and system administrators, this includes:

- **Step-by-step instructions** for key tasks such as:
 - Creating a new lease record
 - Viewing tenant profiles and payment history
 - Generating and exporting reports
 - Updating lease statuses (active, expired, terminated)
- **Common Troubleshooting Tips:**
 - Handling record save errors caused by validation rules
 - What to do when approval processes or flows don't trigger
 - Resolving login issues and access permission problems
- **FAQs Section:**

Answers to frequently asked questions, separated for general users and system administrators. Helps reduce support requests and onboarding time for new users.